Two-stage Stewart platform: ManifoldsX and MuJoCo against forward kinematics

manifoldsx.github.io/stewart-benchmark

A two-stage Stewart platform is driven by its 12 leg lengths for 18 s. With every joint closed, each platform's pose is pure geometry, so the exact answer is known without running any dynamics. Over the full run, ManifoldsX keeps both platforms within 10^{-6} mm (10^{-9} m) in position and 2×10^{-9} rad in orientation of the exact pose, at about 12× real time on one core of a 2.1 GHz Intel Xeon. MuJoCo, with the stiffest constraint settings it accepts, reaches 0.11 mm at 1 ms steps (16×) and 3.1×10^{-3} mm at 0.25 ms (4×); with its default settings the error is 43 mm at any step size. One model and one load case; the limits are listed in section 5.

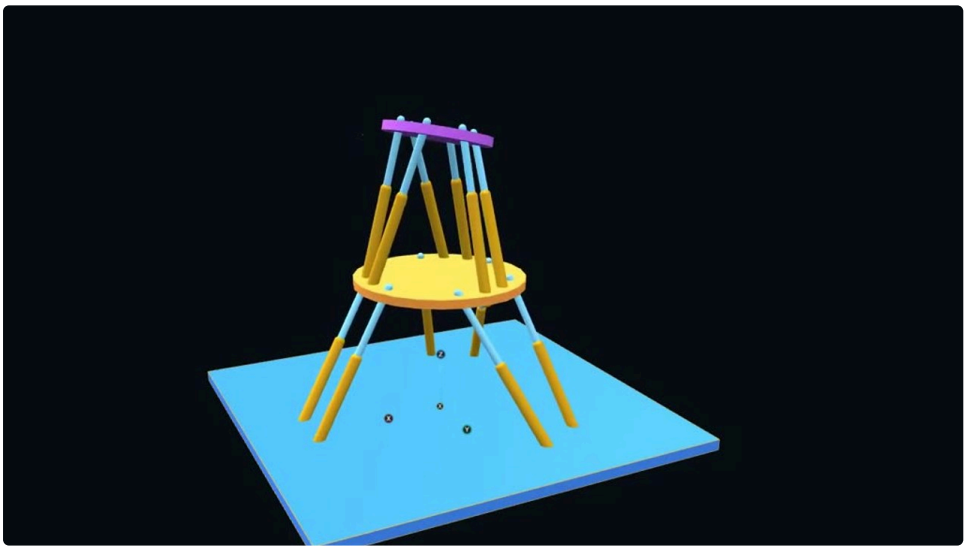


Figure 1. The two-stage Stewart platform: 26 moving bodies driven by 12 leg lengths.

Background

In 1954 Eric Gough, an engineer at the Dunlop tyre works in Birmingham, had a machine running that could load a tyre in several directions at once. Its moving platform stood on six adjustable legs [1]. Eleven years later D. Stewart described a platform with six degrees of freedom for flight simulators [2], and in between the American engineer Klaus Cappel had arrived at the same layout on his own. Six legs between a base and a moving plate became the standard motion base for flight simulators. The same idea now positions telescopes, machine tools and surgical frames.

The layout is also a demanding test for a simulator. Every leg closes a kinematic loop, and stacking two platforms puts 26 moving bodies and 12 loops into one model. Some fast engines, MuJoCo among them [3], treat such constraints as soft: a loop may open slightly, which keeps the solver stable but lets the platform drift from where its legs say it should be. When the leg lengths are

prescribed, the pose of each platform is pure geometry. The right answer is therefore known without running any dynamics, which makes the model a clean way to see how much accuracy a given treatment of loops gives up.

1. Results

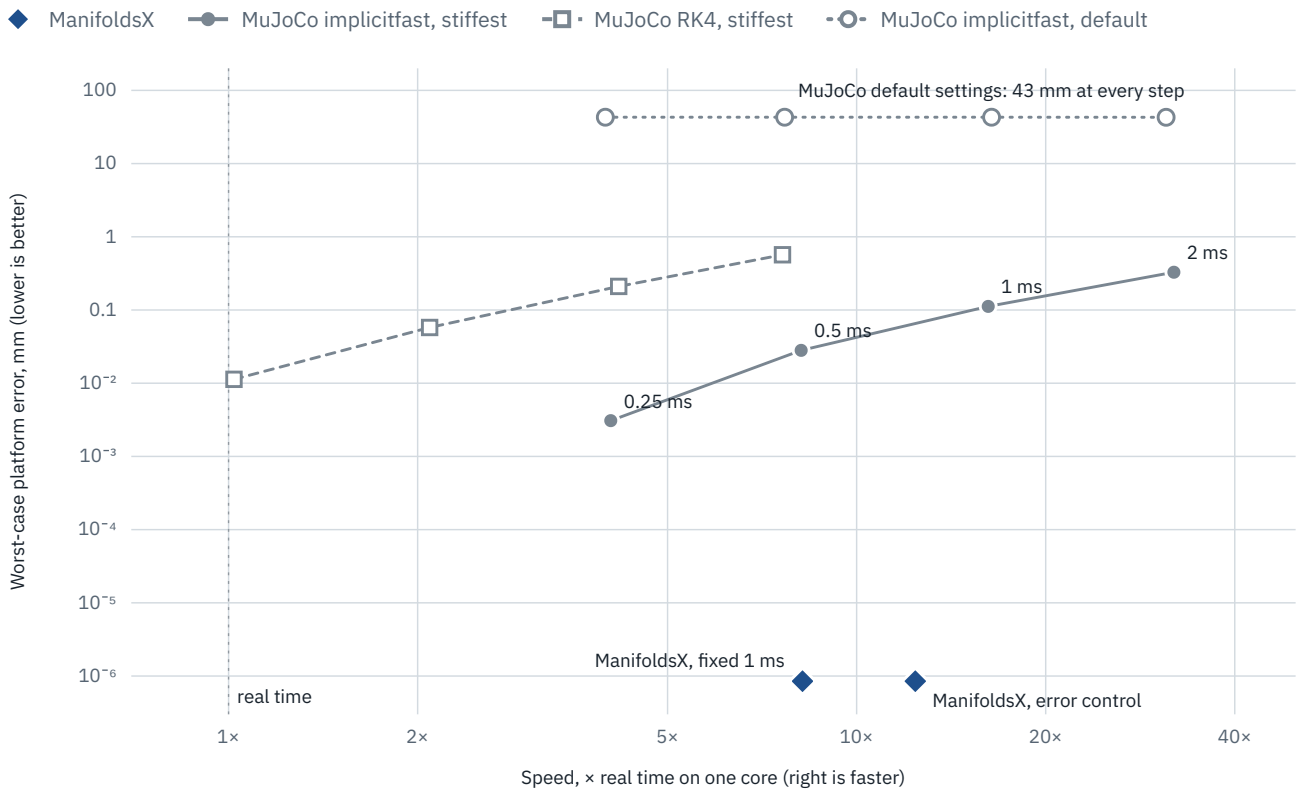


Figure 2. Speed against worst-case top-platform error. Both axes are logarithmic; right is faster, lower is more accurate. Labels on the MuJoCo points are step sizes.

Table 1. Worst values over the 18 s run, one core of a 2.1 GHz Intel Xeon (medians of 3–5 runs)

Setup	Speed (× real time)	Loop gap (mm)	Leg-length error (mm)	Platform error (mm)
ManifoldsX, HHT index-3, error tolerance 10^{-5}, steps up to 3 ms	12	$< 10^{-8}$	exact	$< 10^{-6}$
ManifoldsX, fixed 1 ms steps, the model file's own settings	8	$< 10^{-9}$	exact	$< 10^{-6}$
MuJoCo implicitfast, 2 ms, stiffest	32	6×10^{-3}	0.20	0.33
MuJoCo implicitfast, 1 ms, stiffest	16	1.7×10^{-3}	0.069	0.11
MuJoCo implicitfast, 0.5 ms, stiffest	8.2	4×10^{-4}	0.017	0.028
MuJoCo implicitfast, 0.25 ms, stiffest	4.1	1×10^{-4}	1.6×10^{-3}	3.1×10^{-3}
MuJoCo RK4, 1 ms / 0.25 ms, stiffest	4.2 / 1.0	8×10^{-4} / 6×10^{-5}	0.13 / 7×10^{-3}	0.21 / 0.011
MuJoCo implicitfast, 1 ms, default settings	16	6.0	7.1	43

"Exact": ManifoldsX drives the leg lengths as constraints, so they hold to its Newton tolerance. "<" marks values below the solver tolerance. Stiffest MuJoCo settings: constraint time constant = 2 × step (MuJoCo's lower limit) and solimp 0.9999. Default: 20 ms and 0.9 / 0.95. Platform orientation is scored the same way: both ManifoldsX runs stay within 2×10^{-9} rad of forward kinematics.

2. Method

Model	Two stacked Stewart platforms: 26 moving bodies, 24 spherical and 12 prismatic joints. The 12 leg lengths follow tables (spline-smoothed, sampled every 1 ms, linear between samples) for 18 s. Both solvers read the same model file.
ManifoldsX	Index-3 HHT ($\alpha = 0.3$) with a full Newton matrix and local error control at 10^{-5}, steps up to 3 ms. Joints and leg drives are exact constraints.
MuJoCo 3.15	Same bodies, masses, inertias and joint points. Each loop closes with an equality connect. Each leg follows its table through a joint equality coupled to a virtual clock joint, so the constraint sees the target's velocity as well as its position. A plain C loop around mj_step refreshes the targets every step.
Reference	Forward kinematics. Both stages are solved from the 12 leg lengths with Newton's method at every saved time, to 10^{-14} m, with no dynamics and no solver involved.
Metrics	Worst case over the run of the top platform's position error against the reference, the largest loop gap, and the largest leg-length error.
Cost	Single-core wall time for the 18 s run on one desktop machine.

3. Why the errors differ

ManifoldsX solves joint and drive positions at every step, so its error is set by its Newton tolerance and does not grow with the step size. MuJoCo enforces constraints at acceleration level with stabilization. Each constraint trails by roughly acceleration × time constant^2, and the time constant cannot go below twice the step, so its error falls with the square of the step.

4. Machine

Every speed in this report was measured on one core of a 2-core cloud machine with a 2.1 GHz Intel Xeon processor, as the median of 3–5 runs. Earlier runs on two other machines gave the same errors and speed ratios within about 10 %.

5. What this result does not show

This is one model under prescribed motion. It measures closed-loop kinematic accuracy and says nothing about contact, force-driven dynamics or throughput across many parallel environments, which is what MuJoCo is designed for.

MuJoCo has no built-in time-driven joint constraint, so the clock-joint drive used here is built from standard MuJoCo features. A better MuJoCo setup for driven loops is welcome and will be added.

ManifoldsX itself is not public. The models, the MuJoCo scripts, the forward-kinematics reference and the saved ManifoldsX trajectories are in the public reproduction package. The speeds come from one machine (section 4).

6. Corrections

An earlier internal version of this benchmark put ManifoldsX at 4×10^{-4} mm. That figure came from scoring against ManifoldsX's own 0.25 ms run through a cubic spline, which overshoots at the 1 ms kinks of the piecewise-linear leg tables. Scored against forward kinematics, it was 2×10^{-5} mm.

Later, ManifoldsX began accepting a step only when every constraint meets its tolerance; before, only the average over all constraints had to. Same model and settings, re-run: the error-controlled run went from 2×10^{-5} mm to below the solver tolerance, and the fixed 1 ms run from 9.7×10^{-3} to 1.2×10^{-3} mm. With the current solver the fixed 1 ms run also stays below 10^{-6} mm, at the same speed. The MuJoCo numbers did not change in any of these re-runs.

References

[1] V. E. Gough, “Contribution to discussion of papers on research in automobile stability, control and tyre performance,” *Proc. Automobile Division, Institution of Mechanical Engineers*, pp. 392–394, 1956–57.

[2] D. Stewart, “A platform with six degrees of freedom,” *Proc. Institution of Mechanical Engineers*, vol. 180, part 1, no. 15, pp. 371–386, 1965–66.

[3] E. Todorov, T. Erez and Y. Tassa, “MuJoCo: A physics engine for model-based control,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 5026–5033, 2012.

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